XLS-R patient-bag multiple-instance learning

Saved evaluation complete

Available saved aggregate evaluations are displayed below.

Selected epoch: 31

Method and evaluation protocol

Frozen encoder, patient bags

Audio is converted to mono at 16 kHz. Each recording supplies up to 5 deterministic 5.0000-second waveform windows. Valid encoder frames are mean-pooled within each window, then window embeddings are averaged equally into one recording representation. Padding masks exclude padded waveform samples and encoder frames; the window cap can leave gaps in very long recordings.

A frozen multilingual XLS-R speech encoder provides recording representations. A trainable LayerNorm → Linear(256) → GELU → Dropout(0.3) projection precedes patient-bag aggregation and a three-class linear diagnosis head. All available recording representations belong to that patient's bag.

Variant A uses mean MIL. Variant B uses learned attention MIL. Variant C uses the same attention MIL with train-only precomputed channel variants: each training recording can use its unaugmented base representation or an augmented representation. Validation and test use base representations.

Patient-primary selection

Five-fold CV combines original training and validation patients as the development pool. All recordings from one person stay together; A/B/C share identical folds and original test patients remain excluded. Mean CV patient macro F1 selects A/B/C; ties use lower mean patient loss, then A → B → C. The selected final model is fitted on original training patients; original validation patients select its checkpoint, followed by one original-test evaluation.

Training uses ordinary shuffled patient batches and inverse training-patient class-frequency weights in cross-entropy. Validation loss is unweighted patient cross-entropy. Single-instance recording results are supplemental and do not tune the bag model, select variants, or select checkpoints.

- Seed: 42
- Patient batch size: 8
- Epoch cap: 100
- Learning rate: 0.0010
- Weight decay: 0.0010
- Model revision: 1a640f32ac3e39899438a2931f9924c02f080a54

Original patient partitions

These are original frozen assignments. Five-fold development CV combines original training and validation patients, keeping all original test patients excluded and using the same folds for A/B/C. Final fitting retains the original train / validation / test roles.

Split	Patients	Recordings	NM patients	AD patients	MCI patients
train	68	2,526	34	22	12
val	15	558	7	5	3
test	15	531	7	5	3

Technical source-domain audit

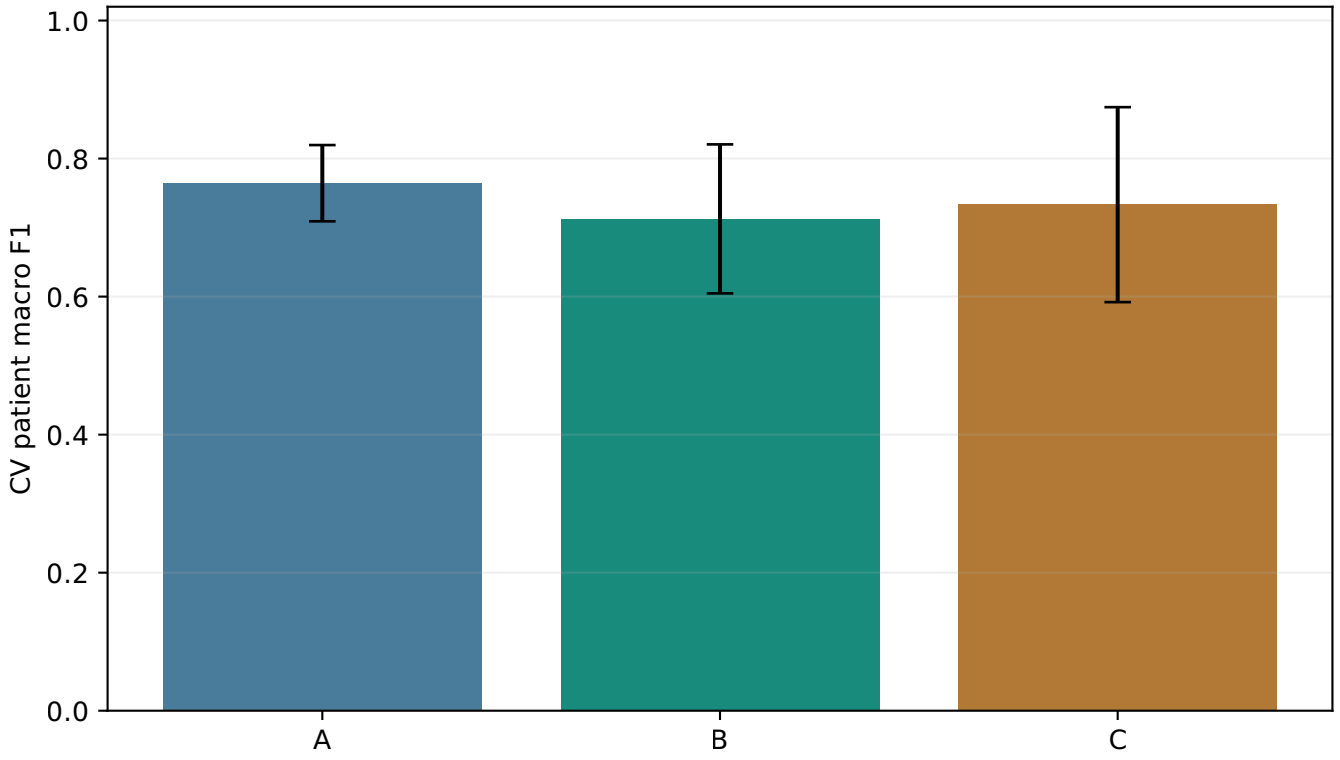
Source recordings: 3,615 · headers inspected: 3,615 · header errors: 0. The audit reads headers across the original train / validation / test partitions; it performs no model evaluation.

Diagnosis	Original split	Recordings	Header errors	Duration min / median / max (s)	Observed rate / channels / encoding: count
NM	train	1,299	0	1.0073 / 3.4392 / 25.1439	12,000 Hz / 1 ch / PCM_16: 38; 96,000 Hz / 1 ch / PCM_16: 1,260; 96,000 Hz / 1 ch / PCM_24: 1
NM	val	261	0	1.1464 / 3.1874 / 18.0529	96,000 Hz / 1 ch / PCM_16: 261
NM	test	279	0	0.9867 / 3.3746 / 16.9537	96,000 Hz / 1 ch / PCM_16: 279
AD	train	753	0	1.1485 / 3.8305 / 57.9868	96,000 Hz / 1 ch / PCM_16: 752; 96,000 Hz / 1 ch / PCM_24: 1
AD	val	179	0	1.1723 / 4.2250 / 55.3055	96,000 Hz / 1 ch / PCM_16: 177; 96,000 Hz / 1 ch / PCM_24: 2
AD	test	133	0	1.0539 / 3.4954 / 36.1342	96,000 Hz / 1 ch / PCM_16: 132; 96,000 Hz / 1 ch / PCM_24: 1
MCI	train	474	0	0.6683 / 3.1334 / 35.5434	16,000 Hz / 1 ch / PCM_16: 474
MCI	val	118	0	0.8662 / 3.1638 / 28.4851	16,000 Hz / 1 ch / PCM_16: 118
MCI	test	119	0	0.8928 / 2.8008 / 19.8853	16,000 Hz / 1 ch / PCM_16: 119

These are available technical metadata, not device or site identities. A diagnosis is never used to infer a recording device. Resampling and channel transformations do not establish domain invariance; clinical labels are not verified by audio headers.

Cross-validation comparison

Selected variant: **A**. Selection uses highest mean CV patient macro F1, then lower mean patient loss, then stable A → B → C order. Original test data does not select the variant.



CV variant means; error bars are the supplied fold standard deviations, not confidence intervals · PNG · SVG

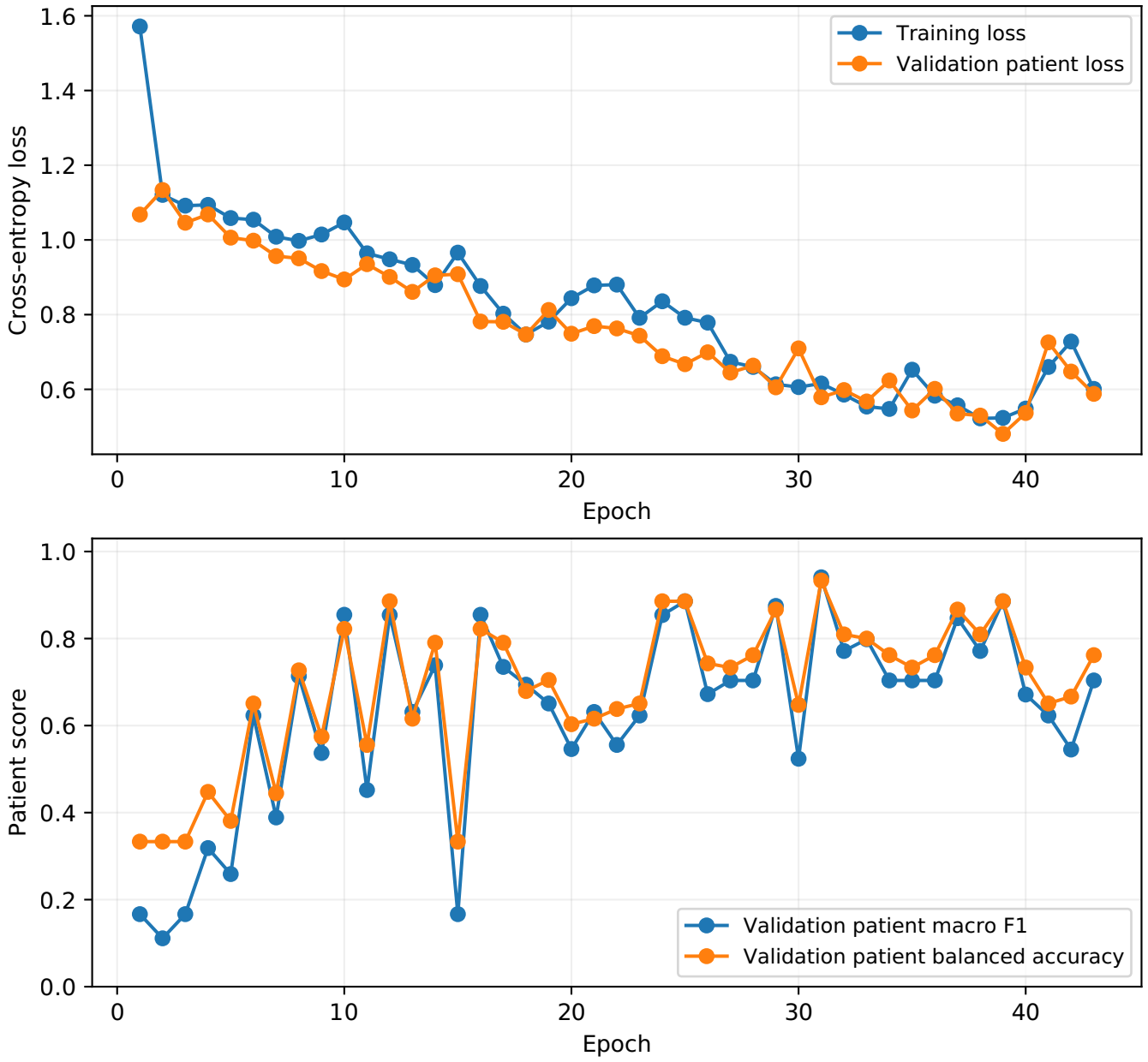
Variant	Saved folds	Macro F1 mean ± SD	Balanced accuracy mean ± SD	Macro AUC mean ± SD	Loss mean ± SD
A	5	0.7643 ± 0.0552	0.7543 ± 0.0572	0.8524 ± 0.0416	0.7245 ± 0.1292
B	5	0.7126 ± 0.1079	0.7209 ± 0.1104	0.8162 ± 0.0789	0.7992 ± 0.2438
C	5	0.7332 ± 0.1412	0.7531 ± 0.1324	0.8563 ± 0.0669	0.8210 ± 0.1991

Saved fold patient metrics

Variant	Fold	Patients	Macro F1	Balanced accuracy	Macro AUC	Loss
A	1	17	0.8363	0.8296	0.9231	0.5401
A	2	17	0.7556	0.7222	0.8567	0.8008
A	3	17	0.7143	0.7222	0.8245	0.8327
A	4	16	0.8045	0.8000	0.8267	0.6373
A	5	16	0.7111	0.6972	0.8309	0.8117
B	1	17	0.6873	0.6519	0.7742	0.9445
B	2	17	0.6528	0.6528	0.7877	0.9696
B	3	17	0.9028	0.9028	0.9566	0.3743
B	4	16	0.6789	0.6472	0.7885	0.8294
B	5	16	0.6413	0.7500	0.7737	0.8779
C	1	17	0.8363	0.8296	0.8380	0.6218
C	2	17	0.9048	0.9167	0.9566	0.6052
C	3	17	0.5558	0.5694	0.8763	0.9685
C	4	16	0.7269	0.7583	0.8363	0.8693
C	5	16	0.6424	0.6917	0.7746	1.0405

Values are read from cv_summary.json. Missing means, deviations or AUCs remain unavailable. Fold histories are never merged into one learning curve.

Learning curves



Training and patient-validation progress for one run or fold · PNG · SVG

Epoch	Train loss	Patient validation loss	Patient macro F1	Patient balanced accuracy
1	1.5715	1.0679	0.1667	0.3333
2	1.1206	1.1335	0.1111	0.3333
3	1.0917	1.0460	0.1667	0.3333
4	1.0938	1.0682	0.3185	0.4476
5	1.0585	1.0061	0.2588	0.3810
6	1.0542	0.9980	0.6231	0.6508
7	1.0086	0.9566	0.3889	0.4444
8	0.9974	0.9507	0.7136	0.7270
9	1.0145	0.9166	0.5369	0.5746
10	1.0467	0.8942	0.8546	0.8222
11	0.9642	0.9351	0.4519	0.5556
12	0.9481	0.9009	0.8540	0.8857
13	0.9330	0.8609	0.6313	0.6159
14	0.8790	0.9052	0.7387	0.7905
15	0.9659	0.9082	0.1667	0.3333
16	0.8766	0.7815	0.8546	0.8222
17	0.8025	0.7808	0.7348	0.7905
18	0.7467	0.7469	0.6940	0.6794
19	0.7812	0.8124	0.6508	0.7048
20	0.8439	0.7489	0.5461	0.6032
21	0.8783	0.7693	0.6313	0.6159
22	0.8803	0.7630	0.5556	0.6381
23	0.7914	0.7435	0.6231	0.6508
24	0.8358	0.6887	0.8540	0.8857
25	0.7916	0.6673	0.8857	0.8857
26	0.7786	0.6991	0.6722	0.7429
27	0.6738	0.6448	0.7037	0.7333
28	0.6600	0.6635	0.7037	0.7619
29	0.6136	0.6054	0.8750	0.8667
30	0.6060	0.7096	0.5238	0.6476
31	0.6157	0.5787	0.9407	0.9333
32	0.5859	0.5981	0.7714	0.8095
33	0.5539	0.5675	0.7983	0.8000
34	0.5476	0.6236	0.7037	0.7619
35	0.6525	0.5434	0.7037	0.7333

Epoch	Train loss	Patient validation loss	Patient macro F1	Patient balanced accuracy
36	0.5829	0.6012	0.7037	0.7619
37	0.5572	0.5350	0.8468	0.8667
38	0.5222	0.5297	0.7714	0.8095
39	0.5236	0.4807	0.8857	0.8857
40	0.5485	0.5366	0.6713	0.7333
41	0.6599	0.7258	0.6231	0.6508
42	0.7280	0.6476	0.5450	0.6667
43	0.6012	0.5879	0.7037	0.7619

Latest completed epoch validation

Patient level · n = Unavailable

Primary assessment: one patient-bag prediction per patient.

Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
70.4%	**76.2%**	**Unavailable**	**Unavailable**	**0.5879**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	Unavailable	Unavailable	Unavailable	Unavailable	Unavailable
AD	Unavailable	Unavailable	Unavailable	Unavailable	Unavailable
MCI	Unavailable	Unavailable	Unavailable	Unavailable	Unavailable

Confusion matrix

No confusion matrix is available.

ROC AUC uses saved class probabilities. Without both positive and negative examples, a class AUC is unavailable; full three-class macro AUC is unavailable if any class AUC is undefined.

Selected checkpoint validation

Patient level · n = 15

Primary assessment: one patient-bag prediction per patient.

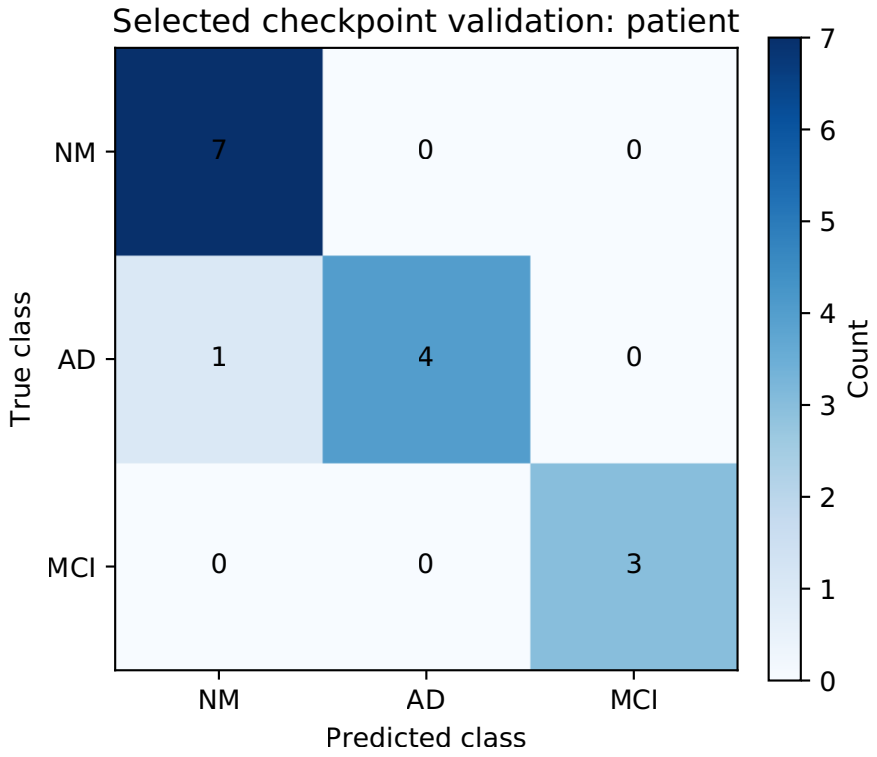
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
94.1%	**93.3%**	**93.3%**	**96.0%**	**0.5787**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	87.5%	100.0%	93.3%	100.0%	7
AD	100.0%	80.0%	88.9%	88.0%	5
MCI	100.0%	100.0%	100.0%	100.0%	3

Confusion matrix

True / predicted	NM	AD	MCI
NM	7	0	0
AD	1	4	0
MCI	0	0	3



Selected checkpoint validation: patient confusion counts · PNG · SVG

Recording level · n = 558

Supplementary single-instance assessment: each recording inherits its patient's diagnosis. Correlated recordings are not additional independent patients and do not select the variant or checkpoint.

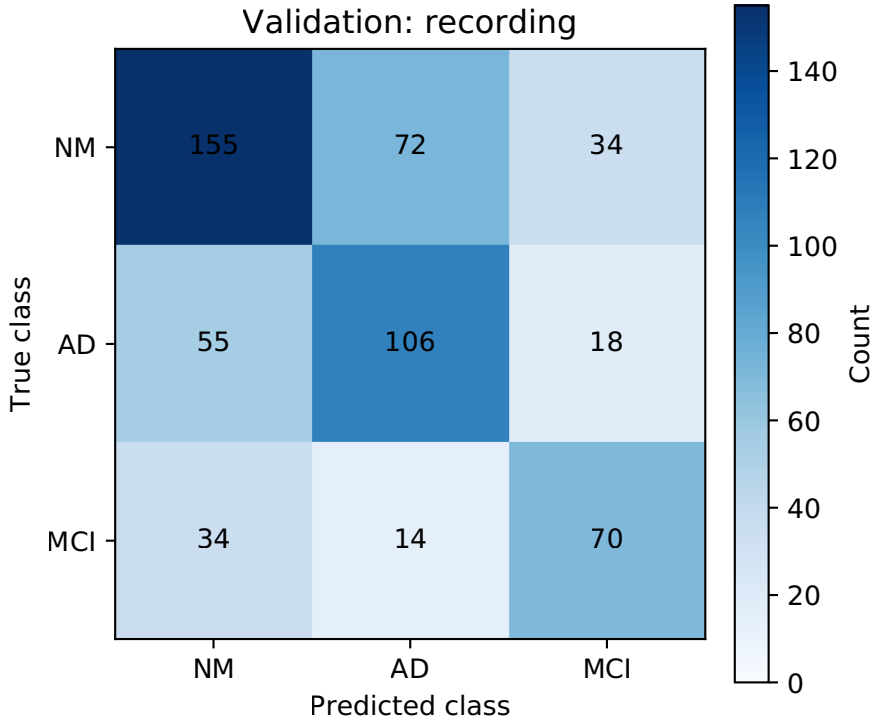
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
59.0%	**59.3%**	**59.3%**	**78.9%**	**0.8900**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	63.5%	59.4%	61.4%	72.6%	261
AD	55.2%	59.2%	57.1%	76.5%	179
MCI	57.4%	59.3%	58.3%	87.6%	118

Confusion matrix

True / predicted	NM	AD	MCI
NM	155	72	34
AD	55	106	18
MCI	34	14	70



Selected checkpoint validation: recording confusion counts · PNG · SVG

ROC AUC uses saved class probabilities. Without both positive and negative examples, a class AUC is unavailable; full three-class macro AUC is unavailable if any class AUC is undefined.

Final test evaluation

Patient level · n = 15

Primary assessment: one patient-bag prediction per patient.

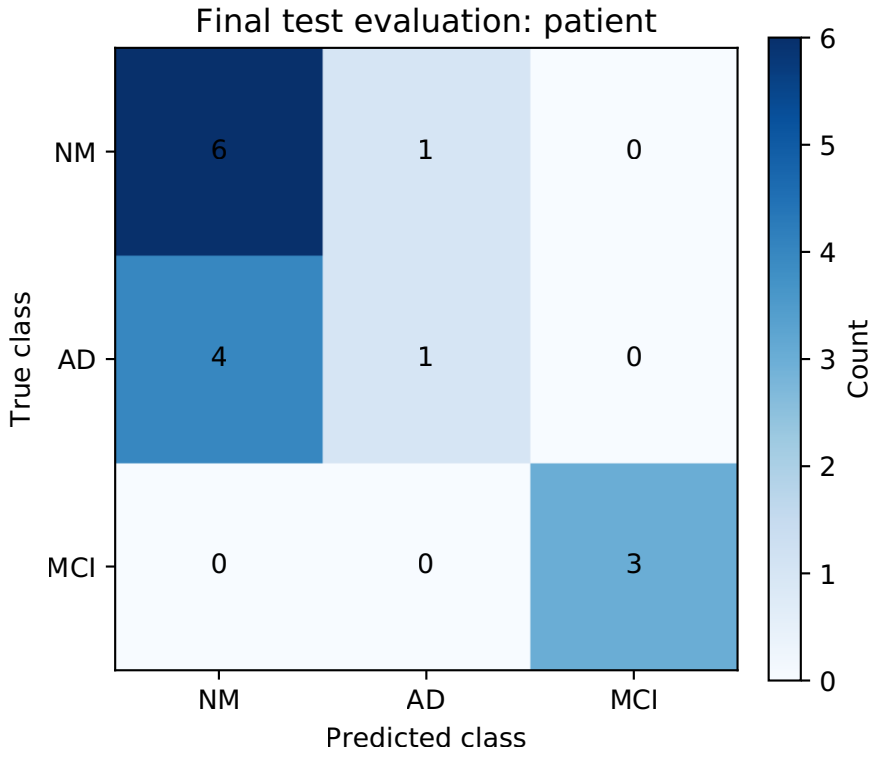
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
66.4%	**68.6%**	**66.7%**	**81.7%**	**0.6306**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	60.0%	85.7%	70.6%	73.2%	7
AD	50.0%	20.0%	28.6%	72.0%	5
MCI	100.0%	100.0%	100.0%	100.0%	3

Confusion matrix

True / predicted	NM	AD	MCI
NM	6	1	0
AD	4	1	0
MCI	0	0	3



Final test evaluation: patient confusion counts · PNG · SVG

Recording level · n = 531

Supplementary single-instance assessment: each recording inherits its patient's diagnosis. Correlated recordings are not additional independent patients and do not select the variant or checkpoint.

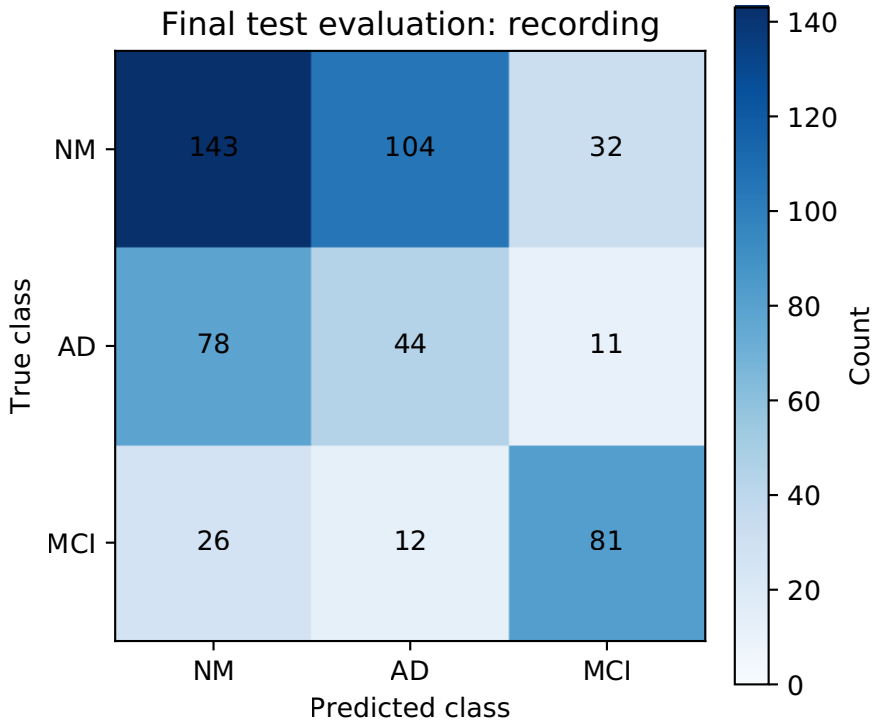
Macro F1	Balanced accuracy	Accuracy	Macro OvR ROC AUC	Loss
50.4%	**50.8%**	**50.5%**	**69.1%**	**0.9579**

Per-class performance

Class	Precision	Recall	F1	OvR ROC AUC	Support
NM	57.9%	51.3%	54.4%	59.8%	279
AD	27.5%	33.1%	30.0%	59.3%	133
MCI	65.3%	68.1%	66.7%	88.1%	119

Confusion matrix

True / predicted	NM	AD	MCI
NM	143	104	32
AD	78	44	11
MCI	26	12	81



Final test evaluation: recording confusion counts · PNG · SVG

ROC AUC uses saved class probabilities. Without both positive and negative examples, a class AUC is unavailable; full three-class macro AUC is unavailable if any class AUC is undefined.

Exploratory s00 / s01 comparison

s00 test results were already known before s01 design. This is reuse of a known holdout, not a new blinded confirmatory comparison. Compare only when private patient memberships match; equal aggregate counts alone do not establish matched membership.

System	Split	Patients	Macro F1	Balanced accuracy	Macro AUC
s00	validation	15	0.8322	0.8381	0.8990
s00	test	15	0.6571	0.6571	0.8486
s01	validation	15	0.9407	0.9333	0.9600
s01	test	15	0.6639	0.6857	0.8174

Interpretation and source confounding

Test class supports can be very small; inspect the per-class patient counts. Recording observations within a patient are correlated. A high class AUC on this holdout does not establish clinical generalization.

s00 test results were already known before s01 design. Comparisons on this reused holdout are exploratory and are not a fresh blinded confirmatory test. Validation and CV results used for model selection are not independent test estimates.

Channel transforms do not establish domain invariance. Cross-device or cross-site generalization requires independent devices or sites with diagnosis overlap. Sample rate, duration and encoding are available technical metadata; diagnosis must never be treated as a measured device identity.

s01 aggregate report · local figures only · no model inference during report generation